**NATIONAL INSTITUTE OF TECHNOLOGY CALICUT**

**Department of Physics | PH7302 Condensed Matter Physics (2026–27)**

**Mid-Semester Evaluation Rubric & Comprehensive Solutions Manual**

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# Evaluation Architecture for the TA

- **Section A (MCQs):** Q1–Q5 are compulsory ($5 \times 1 = 5$ marks). Award full credit (1.0) for the correct option; zero otherwise.
- **Section B (Analytical/Lattice):** Attempt any 3 of Q6–Q9 ($3 \times 8 = 24$ marks). If all 4 are attempted, evaluate all and award marks for the best 3.
- **Section C (Thermal/Transport):** Q10 is compulsory (8 marks). Evaluate any 1 of Q11 or Q12 ($2 \times 8 = 16$ marks).
- **Total Scaled Marks:** 45. Give partial credit exactly as demarcated in the step-by-step mark rubrics below. Do not deduct marks for alternate, mathematically sound notations.

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# SECTION A: MCQs (1 Mark Each)

**Q1.** Plane intercepts: $4a, 2b, \infty c$.
Reciprocals of intercepts:

$$\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{\infty}\right) = \left(\frac{1}{4}, \frac{1}{2}, 0\right)$$

Multiply by the lowest common denominator (4):

$$(1,\ 2,\ 0)$$

**Correct Option: (c)** [1.0 Mark]

**Q2.** Packing fraction of FCC crystal structure.
For hard spheres of radius $R$ in a cubic unit cell with lattice constant $a = 2\sqrt{2}R$ containing $Z = 4$ atoms:

$$\text{APF} = \frac{4 \times \frac{4}{3}\pi R^3}{a^3} = \frac{\frac{16}{3}\pi R^3}{16\sqrt{2}R^3} = \frac{\pi}{3\sqrt{2}} \approx 0.74$$

**Correct Option: (c)** [1.0 Mark]

**Q3.** Monoatomic chain group velocity at Brillouin zone boundary.
Dispersion relation: $\omega(k) = 2\sqrt{C/M}\sin(ka/2)$.
Group velocity:

$$v_g = \frac{d\omega}{dk} = a\sqrt{\frac{C}{M}}\cos\left(\frac{ka}{2}\right)$$

At the zone boundary $k = \pi/a$:

$$v_g\left(\frac{\pi}{a}\right) = a\sqrt{\frac{C}{M}}\cos\left(\frac{\pi}{2}\right) = 0$$

**Correct Option: (c)** [1.0 Mark]

**Q4.** Dulong-Petit high-$T$ molar heat capacity.
Equipartition theorem for 1 mole ($N_A$ atoms) oscillating in 3 spatial dimensions:

$$U = 3N_A k_B T = 3RT \implies C_V = \left(\frac{\partial U}{\partial T}\right)_V = 3R$$

**Correct Option: (a)** [1.0 Mark]

**Q5.** Drude conductivity recovered from the quantum picture.
Due to Pauli exclusion, interior states are blocked. Only electrons in an active window of width $\sim k_B T$ around $E_F$ can gain net momentum, moving at the Fermi velocity $v_F$. The integration over the shifted Fermi surface reproduces $\sigma = ne^2\tau/m$.
**Correct Option: (b)** [1.0 Mark]

# SECTION B (Attempt Any 3)

## Question 6 [Total: 8 Marks]

### (a) Reciprocal Lattice Vectors Definition [2 Marks]

- **Condition [1 Mark]:** $a_i \cdot b_j = 2\pi\delta_{ij}$ (or $\exp(iG\cdot R) = 1$).
- **Explicit Formulas [1 Mark]:**

$$b_1 = 2\pi\frac{a_2 \times a_3}{a_1 \cdot (a_2 \times a_3)}, \quad b_2 = 2\pi\frac{a_3 \times a_1}{a_1 \cdot (a_2 \times a_3)},$$

$$b_3 = 2\pi\frac{a_1 \times a_2}{a_1 \cdot (a_2 \times a_3)}$$

### (b) Orthogonality and Interplanar Spacing [3 Marks]

- **Proof that $G_{hkl} \perp (hkl)$ [1.5 Marks]:**
  Intercepts of plane $(hkl)$ on axes $a_1, a_2, a_3$ are:

$$A = \frac{a_1}{h}, \quad B = \frac{a_2}{k}, \quad C = \frac{a_3}{l}$$

  Vectors lying in the plane include:

$$u = B - A = \frac{a_2}{k} - \frac{a_1}{h}, \quad v = C - B = \frac{a_3}{l} - \frac{a_2}{k}$$

  Taking dot product with $G_{hkl} = hb_1 + kb_2 + lb_3$:

$$G_{hkl} \cdot u = G \cdot \left(\frac{a_2}{k} - \frac{a_1}{h}\right) = \frac{k(2\pi)}{k} - \frac{h(2\pi)}{h} = 0$$

  Similarly, $G_{hkl} \cdot v = 0$. Because $G_{hkl}$ is orthogonal to two linearly independent vectors in $(hkl)$, it is normal to the plane.

- **Derivation of $d_{hkl} = 2\pi/|G_{hkl}|$ [1.5 Marks]:**
  The unit normal is $\hat{n} = G_{hkl}/|G_{hkl}|$. The interplanar spacing $d_{hkl}$ is the projection of any intercept vector (e.g., $A = a_1/h$) onto $\hat{n}$:

$$d_{hkl} = A \cdot \hat{n} = \frac{a_1}{h} \cdot \frac{G_{hkl}}{|G_{hkl}|} = \frac{a_1 \cdot (hb_1 + kb_2 + lb_3)}{h|G_{hkl}|}$$

$$d_{hkl} = \frac{h(2\pi)}{h|G_{hkl}|} = \frac{2\pi}{|G_{hkl}|}$$

**(c) Calculations for Copper (FCC) [3 Marks]**

- **Calculation of $d_{220}$ [1.5 Marks]:**

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

For $a = 3.61$ Å, $(hkl) = (220)$:

$$d_{220} = \frac{3.61 \text{ Å}}{\sqrt{2^2 + 2^2 + 0^2}} = \frac{3.61}{\sqrt{8}} = \frac{3.61}{2.8284} \approx 1.276 \text{ Å}$$

- **Bragg Glancing Angle $\theta$ [1.5 Marks]:**
Bragg's Law: $2d_{220} \sin\theta = n\lambda$ with $n = 1$, $\lambda = 1.54$ Å:

$$\sin\theta = \frac{\lambda}{2d_{220}} = \frac{1.54}{2 \times 1.2763} = \frac{1.54}{2.5526} \approx 0.6033$$

$$\theta = \arcsin(0.6033) \approx 37.1^\circ \quad (\text{or } 0.648 \text{ rad})$$

## Question 7 [Total: 8 Marks]

**(a) Structure Factor Definition & Intensity [2 Marks]**

- **Definition [1 Mark]:** $F_{hkl} = \sum_j f_j \exp[-2\pi i(hx_j + ky_j + lz_j)]$, representing the coherent scattering amplitude from all basis atoms in a single unit cell.
- **Diffraction Intensity [1 Mark]:** Intensity $I_{hkl} \propto |F_{hkl}|^2$. Destructive phase interference occurs when $|F_{hkl}|^2 = 0$, giving systematically absent reflections.

**(b) BCC Structure Factor Derivation [3 Marks]**

- **Derivation [2 Marks]:**
BCC basis: $(0,0,0)$ and $(1/2,1/2,1/2)$ with identical scattering factors $f_j = f$:

$$F_{hkl} = fe^0 + fe^{-2\pi i(h/2+k/2+l/2)} = f\left[1 + e^{-\pi i(h+k+l)}\right]$$

(equivalently $f[1 + e^{\pi i(h+k+l)}]$, since $e^{\pm i\pi N} = (-1)^N$ for integer $N$ — accept either sign convention).

- **Extinction Rule [1 Mark]:**
Using $e^{-\pi iN} = (-1)^N$:

$$F_{hkl} = \begin{cases} 2f, & \text{if } h+k+l \text{ is even (Observed)} \\ 0, & \text{if } h+k+l \text{ is odd (Extinct)} \end{cases}$$

**(c) FCC Structure Factor & Allowed Reflections [3 Marks]**

- **Derivation [1.5 Marks]:**
FCC basis atoms: $(0,0,0)$, $(1/2,1/2,0)$, $(1/2,0,1/2)$, $(0,1/2,1/2)$:

$$F_{hkl} = f\left[1 + e^{-\pi i(h+k)} + e^{-\pi i(h+l)} + e^{-\pi i(k+l)}\right]$$

If indices $h,k,l$ are **unmixed** (all even or all odd), pairwise sums $h+k, h+l, k+l$ are even $\implies F_{hkl} = f[1+1+1+1] = 4f$.
If indices are **mixed** (e.g., two odd, one even), two sums are odd and one is even $\implies F_{hkl} = f[1-1-1+1] = 0$.

- **Reflection Classification [1.5 Marks]:**
Classify the given planes:
  - (100): Mixed $\implies$ **Extinct**
  - (110): Mixed $\implies$ **Extinct**
  - (111): All odd $\implies$ **Observed**
  - (200): All even $\implies$ **Observed**
  - (210): Mixed $\implies$ **Extinct**
  - (220): All even $\implies$ **Observed**

Allowed reflections: **(111), (200), and (220)**.

## Question 8 [Total: 8 Marks]

**(a) Monoatomic Chain Dispersion Derivation [4 Marks]**

- **Equation of Motion [1.5 Marks]:**
Consider harmonic nearest-neighbor restoring forces acting on atom $n$ of mass $M$ at equilibrium position $x_n = na$:

$$M\frac{d^2u_n}{dt^2} = C\left[(u_{n+1} - u_n) - (u_n - u_{n-1})\right]$$

$$M\ddot{u}_n = C[u_{n+1} + u_{n-1} - 2u_n]$$

- **Plane Wave Substitution [1.5 Marks]:**
Assume traveling wave solution $u_n(t) = Ae^{i(kna-\omega t)}$:

$$-M\omega^2 Ae^{i(kna-\omega t)} = CAe^{i(kna-\omega t)}\left[e^{ika} + e^{-ika} - 2\right]$$

$$-M\omega^2 = C[2\cos(ka) - 2] = -2C[1 - \cos(ka)]$$

Using $1 - \cos(ka) = 2\sin^2(ka/2)$:

$$M\omega^2 = 4C\sin^2(ka/2) \implies \omega^2 = \frac{4C}{M}\sin^2(ka/2)$$

- **Final Dispersion Form [1 Mark]:**

$$\omega(k) = 2\sqrt{\frac{C}{M}}\left|\sin\left(\frac{ka}{2}\right)\right|$$

**(b) Physical Quantities [1 Mark]**

- **Speed of sound [0.5 Mark]:** As $k \to 0$, $\sin(ka/2) \approx ka/2$:

$$v_s = \lim_{k\to 0}\frac{\omega(k)}{k} = a\sqrt{\frac{C}{M}}$$

- **Cutoff frequency [0.5 Mark]:** Occurs at zone boundary $k = \pi/a$:

$$\omega_{\max} = 2\sqrt{\frac{C}{M}} \quad \left(\nu_{\max} = \frac{\omega_{\max}}{2\pi}\right)$$

**(c) Numerical Computations [3 Marks]** Given: $M = 40$ amu $= 40 \times 1.66 \times 10^{-27}$ kg $= 6.64 \times 10^{-26}$ kg, $a = 3 \times 10^{-10}$ m, $C = 50$ N/m.

- **Maximum Frequency $\omega_{\max}$ [1 Mark]:**

$$\sqrt{\frac{C}{M}} = \sqrt{\frac{50}{6.64\times10^{-26}}} = \sqrt{7.530\times10^{26}} \approx 2.744\times10^{13} \text{ rad/s}$$

$$\omega_{\max} = 2\sqrt{\frac{C}{M}} \approx 5.488\times10^{13} \text{ rad/s}$$

Converting to ordinary (cyclic) frequency, $\nu_{\max} = \omega_{\max}/2\pi$:

$$\nu_{\max} = \frac{5.488\times10^{13}}{2\pi} \approx 8.73\times10^{12} \text{ Hz } (\approx 8.73 \text{ THz})$$

- **Speed of Sound $v_s$ [1 Mark]:**

$$v_s = a\sqrt{\frac{C}{M}} = (3\times10^{-10} \text{ m})(2.744\times10^{13} \text{ s}^{-1}) \approx 8232 \text{ m/s}$$

- **Group Velocity at $k = 0.3\pi/a$ [1 Mark]:**

$$v_g(k) = \frac{d\omega}{dk} = a\sqrt{\frac{C}{M}}\cos\left(\frac{ka}{2}\right) = v_s\cos\left(\frac{0.3\pi}{2}\right)$$

$$\cos(0.15\pi) = \cos(27^\circ) \approx 0.8910$$

$$v_g = 8232\times0.8910 \approx 7335 \text{ m/s}$$

## Question 9 [Total: 8 Marks]

**(a) Diatomic Lattice Secular Determinant [3 Marks]**

- **Equations of Motion [1 Mark]:**
Unit cell of length $a$ with atoms $M$ at displacement $u_n$ and $m$ at $v_n$:

$$M\ddot{u}_n = C[v_n + v_{n-1} - 2u_n]$$

$$m\ddot{v}_n = C[u_{n+1} + u_n - 2v_n]$$

- **Secular Determinant [1 Mark]:**
Substitute $u_n = ue^{i(kna-\omega t)}$ and $v_n = ve^{i(kna-\omega t)}$:

$$\begin{pmatrix} 2C - M\omega^2 & -C(1+e^{-ika}) \\ -C(1+e^{ika}) & 2C - m\omega^2 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = 0$$

Setting the determinant to zero:

$$(2C - M\omega^2)(2C - m\omega^2) - C^2|1+e^{-ika}|^2 = 0$$

Using $|1+e^{-ika}|^2 = 2+2\cos(ka) = 4\cos^2(ka/2)$, and noting $4\cos^2(ka/2) = 4[1-\sin^2(ka/2)]$, expansion of the determinant gives (*corrected*: the middle term is quadratic in $\omega^2$, not $\omega^4$):

$$Mm\,\omega^4 - 2C(M+m)\,\omega^2 + 4C^2\sin^2(ka/2) = 0$$

- **Dispersion Solution [1 Mark]:**
Solving this quadratic in $\omega^2$ via the quadratic formula:

$$\omega^2(k) = C\left(\frac{1}{M}+\frac{1}{m}\right) \pm C\sqrt{\left(\frac{1}{M}+\frac{1}{m}\right)^2 - \frac{4\sin^2(ka/2)}{Mm}}$$

**(b) Zone-Boundary Band Gap Derivation [2 Marks]** At the first Brillouin zone boundary $k = \pi/a$, $\sin^2(\pi/2) = 1$:

$$\omega^2\left(\frac{\pi}{a}\right) = C\left(\frac{1}{M}+\frac{1}{m}\right) \pm C\sqrt{\left(\frac{1}{m}-\frac{1}{M}\right)^2}$$

$$= C\left(\frac{1}{M}+\frac{1}{m}\right) \pm C\left(\frac{1}{m}-\frac{1}{M}\right)$$

This decouples cleanly:

$$\omega_+^2 = \frac{2C}{m} \implies \omega_+ = \sqrt{\frac{2C}{m}} \quad \text{(Optical Branch)}$$

$$\omega_-^2 = \frac{2C}{M} \implies \omega_- = \sqrt{\frac{2C}{M}} \quad \text{(Acoustic Branch)}$$

The forbidden frequency gap is:

$$\Delta\omega = \omega_+ - \omega_- = \sqrt{\frac{2C}{m}} - \sqrt{\frac{2C}{M}}$$

**(c) Numerical Calculations [3 Marks]** Given: $M = 40$ amu $= 6.64\times10^{-26}$ kg, $m = 20$ amu $= 3.32\times10^{-26}$ kg, $C = 50$ N/m.

- **Acoustic Boundary $\omega_-(\pi/a)$ [1 Mark]:**

$$\omega_- = \sqrt{\frac{2\times50}{6.64\times10^{-26}}} = \sqrt{1.506\times10^{27}} \approx 3.881\times10^{13} \text{ rad/s}$$

$$\nu_- = \frac{\omega_-}{2\pi} \approx \frac{3.881\times10^{13}}{2\pi} \approx 6.18\times10^{12} \text{ Hz } (\approx 6.18 \text{ THz})$$

- **Optical Boundary $\omega_+(\pi/a)$ [1 Mark]:**

$$\omega_+ = \sqrt{\frac{2\times50}{3.32\times10^{-26}}} = \sqrt{3.012\times10^{27}} \approx 5.488\times10^{13} \text{ rad/s}$$

$$\nu_+ = \frac{\omega_+}{2\pi} \approx \frac{5.488\times10^{13}}{2\pi} \approx 8.73\times10^{12} \text{ Hz } (\approx 8.73 \text{ THz})$$

- **Band Gap $\Delta\omega$ [1 Mark]:**

$$\Delta\omega = \omega_+ - \omega_- = (5.488-3.881)\times10^{13} \approx 1.607\times10^{13} \text{ rad/s}$$

$$\Delta\nu = \frac{\Delta\omega}{2\pi} \approx \frac{1.607\times10^{13}}{2\pi} \approx 2.56\times10^{12} \text{ Hz } (\approx 2.56 \text{ THz})$$

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# SECTION C (Q10 Compulsory; Choose Q11 or Q12)

## Question 10 (Compulsory) [Total: 8 Marks]

**(a) 3-D Debye Density of States Derivation [4 Marks]**

- **Mode counting in $k$-space [1.5 Marks]:**
Assuming 3 acoustic modes (1 longitudinal, 2 transverse) with average sound velocity $v_s$, $\omega = v_s k$. Periodic boundary conditions in volume $V$:

$$N(k) = 3\times\frac{\frac{4}{3}\pi k^3}{(2\pi/L)^3} = 3\frac{Vk^3}{6\pi^2} = \frac{Vk^3}{2\pi^2}$$

Expressing in terms of $\omega = v_s k$:

$$N(\omega) = \frac{V\omega^3}{2\pi^2 v_s^3}$$

- **Density of states $g(\omega)$ [1.5 Marks]:**

$$g(\omega) = \frac{dN}{d\omega} = \frac{3V\omega^2}{2\pi^2 v_s^3}$$

- **Debye cutoff condition [1 Mark]:**
Integrating up to $\omega_D$ accounts for all $3N$ normal modes:

$$\int_0^{\omega_D} g(\omega)\, d\omega = \frac{3V}{2\pi^2 v_s^3} \int_0^{\omega_D} \omega^2\, d\omega = \frac{V\omega_D^3}{2\pi^2 v_s^3} = 3N$$

$$\implies \omega_D^3 = 6\pi^2 \frac{N}{V} v_s^3 \implies \omega_D = v_s \left(6\pi^2 \frac{N}{V}\right)^{1/3}$$

Eliminating $v_s^3$ yields:

$$g(\omega) = \frac{9N}{\omega_D^3}\omega^2 \quad (\text{for } 0 \le \omega \le \omega_D)$$

**(b) Low-$T$ Debye $T^3$ Law Derivation [1 Mark]** Internal thermal vibrational energy:

$$U = \int_0^{\omega_D} \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} g(\omega)\, d\omega = \frac{9N\hbar}{\omega_D^3} \int_0^{\omega_D} \frac{\omega^3}{e^{\hbar\omega/k_B T} - 1}\, d\omega$$

Let $x = \frac{\hbar\omega}{k_B T}$ and $\Theta_D = \frac{\hbar\omega_D}{k_B}$. At low temperatures $T \ll \Theta_D$, extend upper limit $x_D = \Theta_D/T \to \infty$:

$$U = \frac{9Nk_B T}{\Theta_D^3} T^3 \int_0^{\infty} \frac{x^3}{e^x - 1}\, dx$$

Standard integral: $\int_0^\infty \frac{x^3}{e^x-1} dx = \frac{\pi^4}{15}$:

$$U = \frac{9Nk_B T^4}{\Theta_D^3} \left(\frac{\pi^4}{15}\right) = \frac{3\pi^4}{5} Nk_B \frac{T^4}{\Theta_D^3}$$

Differentiating with respect to $T$:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{12\pi^4}{5} Nk_B \left(\frac{T}{\Theta_D}\right)^3$$

**(c) Calculations for Sodium [3 Marks]** Given: $\nu_D = 3.3 \times 10^{12}$ Hz (i.e., $\omega_D = 2\pi\nu_D \approx 2.074 \times 10^{13}$ rad/s), $T = 10$ K, $N_A = 6.022 \times 10^{23}$ mol$^{-1}$.

- **Debye Temperature $\Theta_D$ [1.5 Marks]:**

$$\Theta_D = \frac{h\nu_D}{k_B} = \frac{6.626 \times 10^{-34}\ \text{J}\cdot\text{s} \times 3.3 \times 10^{12}\ \text{s}^{-1}}{1.38 \times 10^{-23}\ \text{J/K}}$$

$$\Theta_D = \frac{2.1866 \times 10^{-21}}{1.38 \times 10^{-23}} \approx 158.45\ \text{K}$$

- **Molar Heat Capacity at $T = 10$ K [1.5 Marks]:**
Here $T/\Theta_D = 10/158.45 \approx 0.0631 < 0.1$, so the $T^3$ approximation is valid.

$$C_V = \frac{12\pi^4}{5} R \left(\frac{T}{\Theta_D}\right)^3$$

With $R = N_A k_B = 8.314$ J mol$^{-1}$K$^{-1}$:

$$\frac{12\pi^4}{5} R = 2.4 \times 97.409 \times 8.314 \approx 1943.6\ \text{J mol}^{-1}\text{K}^{-1}$$

$$C_V = 1943.6 \times (0.06311)^3 = 1943.6 \times (2.514 \times 10^{-4})$$

$$C_V \approx 0.489\ \text{J mol}^{-1}\text{K}^{-1}$$

## Question 11 [Total: 8 Marks]

**(a) Free Electron DOS & $E_F$ Derivation [4 Marks]**

- **Density of states derivation [2.5 Marks]:**
In a cubic box of volume $V = L^3$, periodic boundary conditions yield allowed wavevectors $k_i = \frac{2\pi n_i}{L}$. Each mode occupies $k$-space volume $(2\pi/L)^3 = 8\pi^3/V$. Number of electron states (including spin degeneracy $g_s = 2$) with wavevector inside sphere of radius $k$:

$$N(k) = 2 \times \frac{\frac{4}{3}\pi k^3}{(2\pi)^3/V} = \frac{Vk^3}{3\pi^2}$$

Using $E = \frac{\hbar^2 k^2}{2m} \implies k = \left(\frac{2mE}{\hbar^2}\right)^{1/2}$:

$$N(E) = \frac{V}{3\pi^2}\left(\frac{2mE}{\hbar^2}\right)^{3/2} = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{3/2}$$

Taking the derivative:

$$D(E) = \frac{dN}{dE} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E}$$

- **Fermi energy derivation [1.5 Marks]:**
At $T = 0$ K, all $N$ electrons fill states up to $k_F$:

$$N = \frac{Vk_F^3}{3\pi^2} \implies n = \frac{N}{V} = \frac{k_F^3}{3\pi^2} \implies k_F = (3\pi^2 n)^{1/3}$$

$$E_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$$

**(b) Calculations for Sodium [2 Marks]** Given: $\rho = 970$ kg/m$^3$, $A = 23$ g/mol $= 0.023$ kg/mol, monovalent ($z = 1$).

- **Conduction Electron Density $n$ [0.5 Mark]:**

$$n = \frac{\rho N_A}{A} = \frac{970 \times 6.022 \times 10^{23}}{0.023} \approx 2.54 \times 10^{28}\ \text{m}^{-3}$$

- **Fermi Energy $E_F$ [1 Mark]:**

$$3\pi^2 n = 3\pi^2(2.54 \times 10^{28}) = 7.521 \times 10^{29}\ \text{m}^{-3}$$

$$(3\pi^2 n)^{1/3} \approx 9.094 \times 10^9\ \text{m}^{-1}$$

$$E_F = \frac{(1.054 \times 10^{-34})^2}{2 \times 9.11 \times 10^{-31}}(9.094 \times 10^9)^2$$

$$E_F = \frac{1.1109 \times 10^{-68}}{1.822 \times 10^{-30}} \times 8.270 \times 10^{19} \approx 5.04 \times 10^{-19}\ \text{J}$$

In electron-volts:

$$E_F = \frac{5.04 \times 10^{-19}\ \text{J}}{1.602 \times 10^{-19}\ \text{J/eV}} \approx 3.15\ \text{eV}$$

- **Fermi Temperature $T_F$ [0.5 Mark]:**

$$T_F = \frac{E_F}{k_B} = \frac{5.04 \times 10^{-19}\ \text{J}}{1.38 \times 10^{-23}\ \text{J/K}} \approx 36,520\ \text{K}$$

**(c) Ground-State Energy & Active Fraction [2 Marks]**

- **Average Kinetic Energy at $T = 0$ K [1 Mark]:**

$$\langle E\rangle_0 = \frac{U_0}{N} = \frac{1}{N}\int_0^{E_F} E\,D(E)\,dE$$

Since $D(E) = C\sqrt{E}$ and $N = \int_0^{E_F} C\sqrt{E}\,dE = \frac{2}{3}CE_F^{3/2}$:

$$U_0 = C\int_0^{E_F} E^{3/2}\,dE = \frac{2}{5}CE_F^{5/2} \implies \langle E\rangle_0 = \frac{3}{5}E_F$$

$$\langle E\rangle_0 = 0.6 \times 3.15 \text{ eV} \approx 1.89 \text{ eV} \quad (3.02 \times 10^{-19} \text{ J})$$

- **Fraction Excited at $T = 300$ K [1 Mark]:**
Thermal energy range: $\Delta E = 2k_BT$. Conduction electrons excited around $E_F$:

$$f_{\text{th}} \approx \frac{D(E_F) \times 2k_BT}{N}$$

Since $D(E_F) = \frac{3N}{2E_F}$:

$$f_{\text{th}} \approx \frac{3}{2}\frac{2k_BT}{E_F} = 3\frac{k_BT}{E_F} = 3\frac{T}{T_F}$$

$$f_{\text{th}} \approx 3 \times \frac{300 \text{ K}}{36,520 \text{ K}} \approx 3 \times 0.00821 \approx 0.0246 \quad (2.5\%)$$

*(Full credit if student uses simplified scaling $f \approx T/T_F \approx 0.8\%$).*

## Question 12 [Total: 8 Marks]

### (a) Boltzmann Transport Derivation of $\sigma$ [4 Marks]

- **Steady-State Boltzmann Transport Equation [1 Mark]:**
For spatially uniform electron gas under DC field $\mathcal{E} = \mathcal{E}_x\hat{x}$, the relaxation-time approximation gives:

$$\dot{k}\cdot\nabla_k f = -\frac{f - f_0}{\tau}$$

- **Perturbed Distribution $f(k)$ [1.5 Marks]:**
Semiclassical force: $\hbar\dot{k}_x = -e\mathcal{E}_x$. Expanding to first order ($f \approx f_0$ on LHS):

$$\dot{k}_x\frac{\partial f_0}{\partial k_x} = \left(-\frac{e\mathcal{E}_x}{\hbar}\right)\left(\hbar v_x\frac{\partial f_0}{\partial E}\right) = -e\mathcal{E}_x v_x\frac{\partial f_0}{\partial E}$$

$$-e\mathcal{E}_x v_x\frac{\partial f_0}{\partial E} = -\frac{f - f_0}{\tau} \implies f(k) = f_0 + e\tau\mathcal{E}_x v_x\left(-\frac{\partial f_0}{\partial E}\right)$$

- **Current Density and Conductivity [1.5 Marks]:**
Current density along $x$:

$$j_x = -e\int\frac{2\,d^3k}{8\pi^3}v_x f(k)$$

Because $v_x f_0(E)$ is odd in $k_x$, the equilibrium integral vanishes:

$$j_x = \frac{e^2\tau\mathcal{E}_x}{4\pi^3}\int v_x^2\left(-\frac{\partial f_0}{\partial E}\right)d^3k$$

At low temperatures, $-\frac{\partial f_0}{\partial E} \approx \delta(E - E_F)$. For cubic crystals, $\langle v_x^2\rangle = \frac{1}{3}v_F^2$:

$$j_x = \frac{e^2\tau\mathcal{E}_x}{4\pi^3}\left(\frac{1}{3}v_F^2\right)\left(\frac{4\pi k_F^2}{\hbar v_F}\right) = \frac{e^2\tau k_F^3}{3\pi^2 m}\mathcal{E}_x$$

Using $n = k_F^3/(3\pi^2)$, this simplifies to $j_x = \frac{ne^2\tau}{m}\mathcal{E}_x$. Comparing with Ohm's Law $j_x = \sigma\mathcal{E}_x$:

$$\sigma = \frac{ne^2\tau}{m}$$

**(b) Calculations for Copper [2 Marks]** Given: $n = 8.5 \times 10^{28}$ m$^{-3}$, $\tau = 2.5 \times 10^{-14}$ s, $\mathcal{E} = 1$ V/m.

- **Conductivity $\sigma$ [1 Mark]:**

$$\sigma = \frac{(8.5 \times 10^{28})(1.602 \times 10^{-19})^2(2.5 \times 10^{-14})}{9.11 \times 10^{-31}}$$

$$\sigma = \frac{8.5 \times 10^{28} \times 2.566 \times 10^{-38} \times 2.5 \times 10^{-14}}{9.11 \times 10^{-31}}$$

$$\sigma = \frac{5.454 \times 10^{-23}}{9.11 \times 10^{-31}} \approx 5.99 \times 10^7\ \Omega^{-1}\text{m}^{-1}$$

- **Drift Velocity $v_d$ [1 Mark]:**

$$v_d = \mu\mathcal{E} = \frac{e\tau}{m}\mathcal{E} = \frac{(1.602 \times 10^{-19})(2.5 \times 10^{-14})}{9.11 \times 10^{-31}} \times 1$$

$$v_d = \frac{4.005 \times 10^{-33}}{9.11 \times 10^{-31}} \approx 4.40 \times 10^{-3} \text{ m/s} = 4.40 \text{ mm/s}$$

**(c) Fermi Velocity & Mean Free Path [2 Marks]**

- **Fermi Velocity $v_F$ [0.75 Mark]:**

$$k_F = (3\pi^2 n)^{1/3} = (3\pi^2 \times 8.5 \times 10^{28})^{1/3} \approx (2.517 \times 10^{30})^{1/3}$$

$$k_F \approx 1.360 \times 10^{10} \text{ m}^{-1}$$

$$v_F = \frac{\hbar k_F}{m} = \frac{(1.054 \times 10^{-34})(1.360 \times 10^{10})}{9.11 \times 10^{-31}} \approx 1.57 \times 10^6 \text{ m/s}$$

- **Mean Free Path $l$ [0.75 Mark]:**

$$l = v_F\tau = (1.57 \times 10^6 \text{ m/s}) \times (2.5 \times 10^{-14} \text{ s})$$

$$l \approx 3.93 \times 10^{-8} \text{ m} = 39.3 \text{ nm} = 393 \text{ Å}$$

- **Lattice Comparison [0.5 Mark]:**
Ratio relative to interatomic spacing $a = 3.61$ Å:

$$\frac{l}{a} = \frac{393 \text{ Å}}{3.61 \text{ Å}} \approx 109$$

**Physical conclusion:** The electronic mean free path is more than 100 lattice constants long. This demonstrates that electrons traverse hundreds of unit cells without scattering, because an electron wave propagates without resistance through a perfectly periodic crystal potential. Collisions occur only due to lattice defects, impurities, or thermal vibrations (phonons).